EPS-FAILOP-L3

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Fail-Operational Electric Power Steering - L3+ Master System PRD

Automotive Steering Electronics | Rev A1 - Controlled Review Baseline | Architecture, Atomic Requirements, L2 Allocation, Interfaces & Verification

Mission profile. The EPS provides manual steering assist and lateral actuation for L3+ automated driving. The source baseline requires fail-operational lateral control after any single electrical, electronic or sensing fault, with a minimum 50% assist capability available to support safe driver handover or a minimum-risk maneuver. The EPS remains the local authority for motor commutation, electrical protection, thermal limits and steering safety envelopes even when trajectory demand originates in the centralized ADAS controller.

Engineering baseline. The controlled architecture uses a dual-wound synchronous motor, two electrically independent inverter channels, redundant power paths, dual lockstep MCU supervision, dual torque/angle sensing, redundant motor-position feedback and authenticated CAN FD / Ethernet communication. AUTOSAR Classic Platform is the software baseline, while SecOC / MACsec, ASIL-D safety intent, IP6K9K sealing and 15-year lifecycle monitoring are retained from the supplied product definition. Exact vehicle-variant limits that are not explicitly released in the source remain controlled TBCs.

Definition	Controlled product interpretation
Product class	Fail-operational electric power steering for L3+ automated driving and manual steering assist.
Safety / assurance	ASIL-D release objective for manual torque assist, autonomous trajectory execution and prevention of uncommanded steering. A compliance claim is permitted only after the program safety evidence is closed.
Documentation baseline	Program Date TBC - controlled review baseline. External release requires approval of the baseline date and all open release parameters.
Primary architecture	Dual-wound synchronous motor; two redundant inverter/power channels; dual lockstep MCU supervision; dual torque/angle and motor-position sensing; AUTOSAR CP; authenticated CAN FD / Ethernet; local safety-limit authority.
Release philosophy	Runtime faults use bounded safe or degraded behavior. Incomplete safety, DVP, environmental, cybersecurity, manufacturing or supplier evidence blocks production equivalence. Source-defined minimum 50% assist and 15-year lifecycle intent are retained; unsupported numerical limits remain controlled TBCs.

This Master PRD allocates detailed design closure to twelve L2 subsystem PRDs: fail-operational architecture; motor/inverter; power/PMIC; steering sensing; steering control; functional safety; ADAS/EPS interfaces; thermal/mechanical/environmental; EMC/transients; manufacturing/EOL; lifecycle/diagnostics; and DVP/HIL/vehicle validation. Missing analysis, qualification, manufacturing, cybersecurity or supplier evidence blocks the affected variant from release; it is not represented as a fictitious runtime behavior.

PRODUCT GOVERNANCE, BASELINE STATUS & REQUIREMENT QUALITY

The master requirement set separates immutable safety obligations, vehicle-variant calibrations and evidence-dependent release claims. Every steering requirement must have one owner, one measurable intent and an explicit verification route so that the system does not rely on narrative assumptions to close safety-critical behavior.

The source baseline is authoritative for architecture and stated capabilities, including fail-operational behavior, minimum 50% assist after a single fault, redundant sensing/power/control, ASIL-D intent and 15-year lifecycle monitoring. Numerical limits not explicitly released by the program remain TBC and cannot be converted into production acceptance values by editorial inference.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-GOV-001	The EPS master requirement set shall distinguish safety invariants, vehicle-variant calibrations and program-controlled TBC parameters.	Requirements classification audit and TBC register review.
EPS-GOV-002	Each atomic requirement shall trace upward to a mission, hazard-control or interface need and downward to objective analysis or test evidence.	Bidirectional traceability audit.
EPS-GOV-003	Unclosed qualification, manufacturing, cybersecurity, supplier or safety evidence shall prevent production-equivalence approval for the affected configuration.	Release-gate review with open-evidence register.

Verification, margin and future-change rule

Governance is reopened when the product boundary, safety allocation, architecture concept, requirement ownership or baseline date changes. Any revised claim must be reconciled with the L2 subsystem PRDs and the objective evidence matrix before the master baseline advances.

VEHICLE / SYSTEM BOUNDARY & AUTHORITY OWNERSHIP

The EPS receives driver torque, steering-angle information and authenticated trajectory-related commands from the vehicle while owning local motor commutation, current/torque control, thermal derating, actuator diagnostics and enforcement of electrical and mechanical steering limits. Central ADAS demand is therefore an input to EPS authority, not a bypass around it.

The parent vehicle program owns vehicle-level dynamics limits, handover strategy, braking coordination, ODD behavior and MRM orchestration. The EPS owns the correctness and bounded response of the steering actuator inside those allocations. Every interface must identify source, encoding, units, coordinate convention, validity, freshness and timeout behavior.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-SYS-001	The EPS shall reject any external steering demand that exceeds a locally enforced electrical, thermal or mechanical safety limit.	HIL command-limit sweep with local-limit instrumentation.
EPS-SYS-002	Every steering command and status interface shall define source, units, sign convention, freshness, integrity, timeout and authority.	ICD inspection and interface fault injection.
EPS-SYS-003	Loss of an external function shall not remove local EPS protection or motor-control supervision that is required to maintain the allocated steering state.	External-node loss and local-supervision continuity test.

Verification, margin and future-change rule

Boundary changes such as a new ADAS controller, steering rack, ESC coordination concept, power architecture or command representation require an ICD and safety-impact review before the changed interface can be treated as equivalent.

OPERATING MODES, L3 HANDOVER & MRM SUPPORT

The EPS operating-state model includes power-off, initialization, manual assist, L3 autonomous actuation, degraded single-channel operation, MRM support, service/calibration and fault-latched states. Transitions are explicit because steering torque discontinuities during handover can be hazardous even when the target mode itself is valid.

Autonomous-to-manual transfer depends on valid vehicle requests and locally acceptable actuator state. Communication loss, invalid trajectory demand, channel isolation, thermal derating or sensor degradation must drive a deterministic state transition without uncontrolled torque steps or hidden reuse of stale commands.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-MOD-001	Mode transitions shall execute deterministically under maximum communication load, stale-data injection and concurrent diagnostic activity.	HIL state-transition stress campaign.
EPS-MOD-002	Loss of valid ADAS command authority shall transition the EPS to the program-allocated manual or MRM-support state without an uncontrolled steering-torque discontinuity.	Communication-loss and handover dyno test.
EPS-MOD-003	Fault-latched states shall require the released recovery conditions before full steering authority can be restored.	Reset/re-enable negative testing and service-flow review.

Verification, margin and future-change rule

Any change to mode ownership, handover sequencing, MRM interaction, service permissions or fault-latch recovery requires state-transition regression under maximum network and compute load.

L2 SUBSYSTEM ALLOCATION & DOCUMENT INTERFACE MAP

Detailed engineering closure is split into twelve L2 documents so that each physical mechanism has an accountable owner: EPS-01 fail-operational architecture; EPS-02 motor/inverter; EPS-03 power and PMIC; EPS-04 sensing/metrology; EPS-05 steering control; EPS-06 functional safety; EPS-07 ADAS/EPS communication; EPS-08 thermal/mechanical/environmental; EPS-09 EMC/transients; EPS-10 manufacturing/EOL; EPS-11 lifecycle/diagnostics; EPS-12 DVP/HIL/vehicle validation.

The Master PRD owns cross-subsystem intent and conflict resolution. An L2 PRD may refine topology, calibration, timing or acceptance criteria, but it cannot weaken a master safety invariant. Shared values such as current limits, torque capability, FTTI allocations and thermal derates must have one configuration-controlled source of truth rather than copies that can diverge.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-ALLOC-001	Every master requirement shall identify the L2 subsystem document responsible for detailed design and verification closure.	Allocation matrix audit.
EPS-ALLOC-002	A parameter shared by more than one L2 subsystem shall have one declared owner and controlled distribution mechanism.	Parameter ownership and duplicate-definition review.
EPS-ALLOC-003	An L2 requirement shall not relax a master safety or fail-operational obligation without approved change control at the Master PRD level.	Requirements consistency and change-control audit.

Verification, margin and future-change rule

Creating, merging or splitting an L2 document requires an allocation review of requirement IDs, parameter ownership, interfaces and verification evidence so traceability remains continuous.

FAIL-OPERATIONAL CONCEPT & MINIMUM 50% ASSIST MISSION

The source mission requires lateral-control capability to remain available following any single electrical, electronic or sensing fault, with at least 50% of rated assist available to support handover or an MRM. This is a system capability claim and therefore depends on the surviving motor winding, inverter, power path, controller, sensing set and mechanical path remaining mutually compatible after isolation of the faulted channel.

Fail-operational behavior is not equivalent to continuing full performance through every fault. The released degraded envelope must identify surviving torque, steering rate, thermal duration, speed/ODD restrictions and driver/ADAS status. These additional limits remain program-controlled until allocated by the vehicle safety concept.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-FOA-001	Following any single fault within the allocated fail-operational scope, the EPS shall retain at least 50% rated assist capability unless the vehicle safety concept explicitly allocates a more restrictive safe state for that fault.	Single-fault matrix, dyno torque test and safety-case trace.
EPS-FOA-002	The surviving channel shall operate without dependence on a failed channel resource whose loss would defeat the allocated degraded steering function.	Dependency analysis and injected channel-loss test.
EPS-FOA-003	Degraded steering capability shall be reported to the vehicle with explicit availability and limitation status.	Network status verification during single-channel operation.

Verification, margin and future-change rule

A change to redundancy architecture, winding design, power source, isolation mechanism or degraded-control strategy reopens the fail-operational capability proof and associated vehicle validation.

DUAL-WOUND MOTOR ARCHITECTURE & CHANNEL INDEPENDENCE

The electromechanical actuator uses a dual-wound synchronous motor so that one electrical channel can be isolated while the other continues producing steering torque. Exact winding arrangement, phase count, magnetic coupling and neutral strategy are L2 design parameters; the Master PRD controls the independence and surviving-assist objectives.

Electrical faults can couple through the shared rotor, stator iron, housing or mechanical load even when windings are separately driven. The motor design therefore requires analysis of inter-winding insulation, mutual inductance, thermal coupling and fault torque so a shorted or open winding does not create a mechanical lock or excessive opposing torque.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-MOT-001	The two motor winding channels shall be electrically isolatable such that a fault in one winding or inverter does not disable the surviving torque path.	Winding/inverter isolation fault test and insulation review.
EPS-MOT-002	A faulted winding condition shall not create blocking or reverse torque that prevents the surviving channel from meeting the allocated degraded-assist requirement.	Dyno fault emulation and torque-speed characterization.
EPS-MOT-003	Motor thermal limits shall account for single-channel operation because the surviving winding may carry a different loss distribution than in normal operation.	Thermal model and single-channel hot test.

Verification, margin and future-change rule

Changes to winding geometry, insulation system, magnet material, resolver target, motor supplier or rack ratio require renewed electromagnetic, thermal and degraded-mode correlation.

DUAL INVERTER POWER STAGE & FAULT CONTAINMENT

Two inverter channels drive the dual-wound motor. Power semiconductor, DC-link, gate-drive and phase-isolation design must contain short circuit, open phase, stuck gate, desaturation, supply collapse and unintended conduction without propagating the fault into the surviving channel or the mechanical actuator.

The Master PRD does not prescribe a semiconductor technology or gate-driver part number. Those choices belong in EPS-02, together with switching frequency, current rating, short-circuit withstand, dead time and hardware trip thresholds. What is fixed here is the need for bounded fault energy, independent shutdown and compatibility with the fail-operational mission.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-INV-001	Each inverter channel shall provide an independent means to remove gate authority when a local power-stage fault is detected.	Gate-disable timing and processor-stall fault test.
EPS-INV-002	A shorted or uncontrolled inverter leg shall be isolated or otherwise contained before it can violate the allocated uncommanded-steering safety limit.	Power-stage fault injection with torque measurement.
EPS-INV-003	The surviving inverter shall remain electrically operable after isolation of the faulted channel within the released degraded envelope.	Dual-channel fault-containment dyno test.

Verification, margin and future-change rule

A change in MOSFET technology, gate driver, DC-link network, current path, phase isolation or inverter PCB layout requires renewed short-circuit, thermal, EMC and degraded-operation evidence.

REDUNDANT MCU, LOCKSTEP SUPERVISION & FREEDOM FROM INTERFERENCE

Dual lockstep MCU supervision provides deterministic motor-control and safety execution with cross-channel state comparison. AUTOSAR Classic Platform is the source-defined software baseline. Safety-critical control, diagnostics, communication and service functions must be scheduled so lower-criticality work cannot starve commutation or protection timing.

Cross-channel synchronization is useful only when the channels retain sufficient independence to avoid common-mode corruption. Clock, reset, memory, watchdog and inter-MCU communication dependencies therefore require explicit ownership and diagnostic coverage. A disagreement must be classified before it can cause an uncontrolled switch of steering authority.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-MCU-001	Each MCU shall retain measured execution-time and scheduling margin under maximum trajectory, communication and diagnostic load.	WCET analysis and processor-load stress test.
EPS-MCU-002	Cross-channel synchronization or result disagreement shall be detected and contained within the allocated fault-response time.	State-desynchronization and data-corruption injection.
EPS-MCU-003	Non-control software shall not obtain memory, CPU or peripheral access that can delay or corrupt the steering-control safety path.	FFI analysis and resource-exhaustion negative test.

Verification, margin and future-change rule

MCU family, compiler, AUTOSAR stack, scheduler, memory map, watchdog or inter-MCU protocol changes require WCET, FFI, reset and cross-channel supervision regression.

12 V / 48 V REDUNDANT POWER & CROSS-FEED PREVENTION

The source baseline allows redundant 12 V / 48 V power architectures. The released vehicle variant must define independent feeds, grounding, protection, connector ownership and any source-transfer logic. The fail-operational concept requires one power-domain fault not to collapse the surviving control and inverter path.

Common points such as chassis ground, battery distribution, EMI filters or transient clamps can defeat apparent redundancy if not analyzed. Cross-feed, reverse current, load-dump coupling and fault current must therefore be bounded by hardware architecture rather than assumed from schematic separation alone.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-PWR-001	A fault or brownout on one released EPS power channel shall not reset or corrupt the surviving channel required for degraded steering.	Independent-source transient sweep and reset monitoring.
EPS-PWR-002	The power architecture shall prevent a failed channel from back-feeding destructive current into the surviving channel.	Cross-feed analysis and forced reverse-current test.
EPS-PWR-003	Power-source status shall be available to the steering state machine with validity and diagnostic coverage sufficient for controlled degradation.	Power-status fault injection and state-transition test.

Verification, margin and future-change rule

A vehicle power architecture, connector, protection device, grounding scheme or source-ORing change reopens transient, cross-feed and surviving-channel verification.

PMIC, RESET, BROWNOUT & STARTUP / SHUTDOWN SEQUENCING

Independent PMIC and reset domains sequence MCU, sensor and gate-driver supplies into a known state. Startup must prevent uncontrolled gate activity before current sensing, rotor position and safety supervision are valid; shutdown must remove actuation authority while preserving required diagnostic/event data.

Brownout behavior is especially important during cranking, source transfer or damaged harness conditions. An MCU that continues executing below characterized voltage, or a gate driver that enters an undefined UVLO region, can create inconsistent channel states. Thresholds and hysteresis are therefore controlled design parameters in the L2 power PRD.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-PMIC-001	Gate-enable authority shall remain inhibited until the released power-good, reset, sensing and controller-initialization conditions are valid.	Slow-ramp and interrupted-startup test.
EPS-PMIC-002	Brownout detection shall move each affected channel to its defined non-hazardous state before execution or gate behavior becomes indeterminate.	Programmable supply sweep with gate/MCU trace.
EPS-PMIC-003	Shutdown shall preserve the event data required to diagnose a steering fault without prolonging unsafe actuation.	Power-removal and NVM-consistency test.

Verification, margin and future-change rule

Changes to PMIC, reset supervisor, UVLO thresholds, hold-up capacitance, gate-driver supply or startup software require power-cycle and brownout regression.

STEERING TORQUE & ANGLE SENSING (TAS) METROLOGY

Driver-intent and steering-position information use redundant torque/angle sensing with diversity or independent plausibility as defined by the L2 sensing PRD. The source baseline cites Hall and MR as possible diverse technologies; the released design must control the actual sensor principle, channel mapping and calibration.

Safety depends on error bounds, not only nominal accuracy. Offset, gain, nonlinearity, temperature coefficient, hysteresis, latency, saturation, stuck output, supply sensitivity, connector faults and magnetic interference must be covered across beginning and end of life. Common-cause faults require explicit treatment.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-TAS-001	TAS measurement error and latency shall remain within the released steering-control and safety budgets across the approved thermal, tolerance and aging envelope.	Metrology WCCA and environmental bench sweep.
EPS-TAS-002	Stale, saturated, shorted, open or implausible TAS data shall be detected before it can drive an unsafe torque command.	Sensor electrical and data fault injection.
EPS-TAS-003	The sensing architecture shall address common-cause faults through diversity or independently validated plausibility measures.	Common-cause analysis and correlated-fault testing.

Verification, margin and future-change rule

Sensor technology, magnet/target geometry, supply, ADC path, connector, mechanical preload or calibration method changes reopen metrology WCCA and plausibility validation.

MOTOR POSITION SENSING & COMMUTATION INTEGRITY

Rotor-position information is essential to field-oriented commutation. The released variant may use resolver, inductive or another approved sensing principle, but it must detect offset, phase error, loss of excitation, stuck angle, excessive noise and plausibility disagreement before incorrect commutation creates reverse or blocking torque.

A redundant sensor or qualified sensorless fallback may support degraded operation, but the fallback must have an explicitly bounded operating envelope. Low speed, startup and high-torque conditions are particularly important because observability can differ from normal driving.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-POS-001	Rotor-position error shall be detected and contained before it can produce uncommanded torque beyond the allocated safety limit.	Offset, phase and noise injection on motor dyno.
EPS-POS-002	Loss of the primary rotor-position path shall transition to the approved redundant or fallback strategy without mechanical lock.	Sensor-loss dyno campaign across speed/torque corners.
EPS-POS-003	Position validity shall gate torque authority during startup and recovery from a sensing fault.	Startup/recovery negative tests with invalid position data.

Verification, margin and future-change rule

Position-sensor technology, excitation circuit, target geometry, estimator, motor electrical parameters or sensorless fallback changes require renewed commutation and fault-response correlation.

PHASE CURRENT SENSING, LEG DIAGNOSTICS & PLAUSIBILITY

Each inverter requires current information for torque control and protection. The detailed architecture may use shunt, Hall, fluxgate or other approved sensing, but current channels must support the control bandwidth, diagnostic range and fault-response needs of both normal and degraded operation.

Current-sensor offset, saturation or common-mode disturbance can produce incorrect torque even when rotor position is valid. Plausibility therefore compares phase currents, DC-link behavior, commanded voltage and motor model expectations. Protection-grade current detection may be separate from control-grade measurement where independence is required.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-CUR-001	Control-current measurements shall meet the released accuracy, bandwidth and latency budgets across the inverter operating envelope.	Current-injection calibration and dynamic bandwidth test.
EPS-CUR-002	Implausible phase-current combinations shall be detected and prevented from sustaining unsafe torque authority.	Open/short/saturation fault injection.
EPS-CUR-003	Protection-current detection shall remain effective when the primary control software is stalled or corrupted where independence is allocated.	Processor-stall overcurrent test and hardware-path timing trace.

Verification, margin and future-change rule

Current-sensor technology, shunt value, amplifier, ADC reference, blanking, PWM strategy or phase topology changes reopen accuracy, bandwidth and protection correlation.

FIELD-ORIENTED CURRENT LOOP & TORQUE CONTROL

The inner control structure regulates d/q current and commanded motor torque with deterministic timing. Controller bandwidth, PWM frequency, current sampling, dead time and motor parameterization are L2 design parameters; the Master PRD controls stability, bounded saturation and safety-limit precedence.

Fail-operational operation can change available voltage, current and thermal margin. The control law must therefore maintain predictable behavior when only one winding/channel remains available, rather than reusing nominal gains without evidence. Anti-windup and current limiting must prevent recovery overshoot.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-FOC-001	The current control loop shall remain stable across the released electrical, thermal and motor-parameter tolerance envelope.	Frequency-response and step-response correlation.
EPS-FOC-002	Torque/current limiting shall take precedence over external assist or trajectory demand and shall avoid integrator windup that causes recovery overshoot.	Saturation and recovery dyno test.
EPS-FOC-003	Single-channel control shall use gains and limits demonstrated for the degraded electrical plant.	Single-channel stability and torque tracking test.

Verification, margin and future-change rule

Motor parameters, current-loop rate, PWM strategy, estimator, calibration or inverter topology changes require renewed loop-stability and torque-correlation testing.

STEERING ANGLE / POSITION CONTROL & TRAJECTORY TRACKING

In L3 autonomous operation the EPS executes a validated steering demand from the ADAS path while retaining local actuator limits. Depending on the vehicle interface, demand may be represented as angle, curvature, torque or another controlled command. The EPS must convert that representation into bounded motor torque without assuming unlimited rack rate or actuator authority.

Tracking performance is meaningful only inside the released mechanical and thermal envelope. Error handling must distinguish unavailable actuation from a merely large tracking error, and the vehicle must receive enough status to decide whether continued automation, handover or MRM is appropriate.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-TRJ-001	Autonomous steering demand shall be rate- and magnitude-limited by locally enforced actuator and safety constraints.	HIL trajectory sweep and saturation test.
EPS-TRJ-002	Tracking error beyond the released tolerance shall produce explicit degraded/invalid status rather than silent continued authority.	Closed-loop tracking fault test.
EPS-TRJ-003	Trajectory execution shall not depend on stale commands beyond the defined freshness window.	Stale-frame replay and timeout testing.

Verification, margin and future-change rule

Changes to command representation, rack ratio, control bandwidth, vehicle-dynamics interface or trajectory filter require closed-loop tracking and safety-limit regression.

MANUAL ASSIST, AUTONOMOUS BLENDING & STEERING FEEL

Manual driver assist and autonomous steering share the same mechanical actuator, so authority blending must avoid surprise torque and preserve driver controllability. The system must know whether the driver, automation or a degraded controller is commanding torque, and transitions must be monotonic and traceable.

Steering feel functions such as damping, returnability and boost calibration are comfort/performance features but can interact with safety if they create unexpected torque. Their authority is therefore subordinate to local safety limits, driver override logic and the released handover concept.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-BLEND-001	Transition between manual assist and autonomous actuation shall not create a discontinuous steering-torque step beyond the released limit.	Mode-transition torque-ripple measurement.
EPS-BLEND-002	Driver override behavior shall remain available according to the vehicle handover concept and shall be diagnosable if inhibited by a fault.	Driver-torque overlay and fault test.
EPS-BLEND-003	Comfort/feel functions shall not override safety torque, thermal derating or degraded-mode limits.	Priority/arbitration negative testing.

Verification, margin and future-change rule

Assist maps, driver-override thresholds, blending filters, handover timing or steering-feel calibrations require dyno and vehicle regression before release.

FRICTION, INERTIA COMPENSATION & PLANT ADAPTATION

Rack friction, motor inertia, seals, tires and mechanical wear influence steering response. Compensation improves feel and tracking but can become destabilizing if plant parameters drift or if a calibration derived for one rack variant is reused on another. Compensation values are therefore variant-controlled.

The source lifecycle concept includes friction-wear estimation. That estimator can support maintenance and calibration supervision, but it cannot silently increase control authority to mask mechanical binding. A growing friction estimate must be bounded and surfaced as a diagnostic/lifecycle condition.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-FRIC-001	Friction and inertia compensation shall remain bounded by the approved plant model and shall not conceal a mechanical fault condition.	Dyno friction sweep and binding-fault injection.
EPS-FRIC-002	Adaptive or learned compensation shall be constrained to a released parameter range with explicit reset and diagnostic behavior.	Adaptation-boundary and persistence testing.
EPS-FRIC-003	Friction-growth monitoring shall report persistent mechanical degradation without automatically expanding steering torque limits.	Accelerated wear and diagnostic correlation test.

Verification, margin and future-change rule

Rack, motor, gear ratio, lubrication, tire assumptions or compensation algorithm changes reopen plant identification, stability and wear-estimator correlation.

CONTROL STABILITY, SATURATION & LIMIT MANAGEMENT

Current, torque, velocity and position loops form a nested control system whose stability depends on electrical plant, rack dynamics, sensor delay and command filtering. Gain and phase margins must be demonstrated across released tolerances rather than only at nominal temperature and supply voltage.

Limit interactions are part of the stability problem. Voltage saturation, current limit, thermal derating, rack-end protection and degraded single-channel operation can each change available control authority. The controller must enter and exit these limits without oscillation, limit cycling or delayed torque release.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-CTL-001	The closed-loop steering controller shall maintain released stability margins across the approved electrical and mechanical plant envelope.	Frequency-response identification and robustness analysis.
EPS-CTL-002	Entry to and recovery from voltage, current, thermal or mechanical saturation shall not create sustained oscillation or unsafe torque overshoot.	Multi-limit dyno stress test.
EPS-CTL-003	Rack-end and mechanical-travel protection shall override external command demand while preserving controlled driver feedback.	Rack-end approach and fault test.

Verification, margin and future-change rule

Any change in control-loop rate, filter, plant model, saturation strategy, derating law or rack-end logic requires stability and limit-interaction regression.

ASIL-D SAFETY INTENT, SAFETY GOALS & ITEM ASSUMPTIONS

The source assigns ASIL-D intent to manual torque assist, autonomous trajectory execution and prevention of uncommanded steering. The final item definition, hazard analysis, safety goals and allocation belong to the program safety lifecycle; this Master PRD records the engineering obligations that those artifacts must ultimately support.

Safety claims must distinguish loss of assist, excessive assist, unintended torque, blocked steering, delayed torque removal, incorrect direction and unavailable degraded channel. External vehicle mitigations can be credited only when their timing, independence and interface behavior are controlled.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-SAF-001	Every safety-relevant EPS function shall trace to the approved vehicle/item safety concept and its allocated integrity objective.	Safety traceability and item-definition review.
EPS-SAF-002	Safety analysis shall explicitly cover unintended steering, loss/excessive assist, blocked steering and failure of degraded-channel availability.	Hazard/safety-analysis completeness audit.
EPS-SAF-003	External mitigations shall not be credited unless their required timing, independence and interface assumptions are verified.	Safety-interface and dependency review.

Verification, margin and future-change rule

A change to item boundary, vehicle assumptions, safety goal, ASIL allocation, external mitigation or steering authority reopens the safety trace and affected DVP.

FTTI ALLOCATION & END-TO-END FAULT RESPONSE

Fault Tolerant Time Interval is an end-to-end property from physical fault onset to a bounded steering state. The source mentions steering FTTI values typically below 50 ms as context, but this Master PRD does not convert that example into a released number. The vehicle program must allocate and approve the actual timing budgets.

Detection blanking, sensor conversion, software execution, cross-channel communication, gate disable, current decay and electromechanical torque decay all consume the FTTI. Measuring only software reaction time can therefore give a false safety margin.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-FTTI-001	The released FTTI budget shall include detection, processing, communication, gate action, electrical current decay and mechanical torque response.	End-to-end timing budget and oscilloscope/HIL trace.
EPS-FTTI-002	Each safety mechanism shall demonstrate worst-case response time against its allocated portion of the approved FTTI.	Mechanism-specific fault injection at worst-case conditions.
EPS-FTTI-003	The program shall not claim FTTI closure using nominal or average execution timing where worst-case timing is safety relevant.	Timing-evidence audit and WCET review.

Verification, margin and future-change rule

Changes to fault detector, scheduler, network path, gate driver, current decay path or motor mechanics require timing re-budget and end-to-end fault injection.

INDEPENDENT HARDWARE PROTECTIONS & UNCOMMANDED TORQUE PREVENTION

Uncommanded steering prevention cannot rely solely on the same software task that may be faulty. The design therefore allocates independent or sufficiently separated mechanisms such as hardware gate disable, current-limit comparators, watchdog-supervised gate authority and phase-isolation functions where required by the safety concept.

Protection must be designed around credible fault energy and torque response, not a generic number of PWM cycles. Threshold, blanking and turn-off behavior are L2 parameters derived from semiconductor, motor and vehicle safety limits. Nuisance trips also matter because unnecessary loss of assist can create a different hazard.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-PROT-001	At least one allocated uncommanded-torque protection path shall remain effective when the primary application software is stalled.	Processor-stall plus overcurrent/uncommanded-torque test.
EPS-PROT-002	Protection thresholds and blanking shall be derived from released device/fault-energy and steering-safety constraints.	Protection WCCA and hardware-injection test.
EPS-PROT-003	A protection fault or nuisance trigger shall be diagnosed sufficiently to support the allocated degraded-state decision.	Protection stuck-on/stuck-off diagnostic test.

Verification, margin and future-change rule

Power-stage, comparator, gate-driver, watchdog or isolation-topology changes require renewed independence, timing and false-trip validation.

SINGLE-POINT, LATENT & COMMON-CAUSE FAULT CONTAINMENT

Redundancy is credible only when a single fault cannot defeat both channels and latent faults are detected before a second fault becomes hazardous. Shared clocks, grounds, connectors, housings, power feeds, calibration data and communication links require common-cause scrutiny even when the logical architecture appears duplicated.

Latent diagnostic intervals and startup tests are program-controlled safety parameters. The Master PRD requires that they be derived from the approved safety concept and verified using realistic fault injection, including faults that produce plausible but incorrect sensor or controller outputs.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-CCF-001	The released architecture shall identify and control common resources whose failure could disable both steering channels.	Common-cause analysis and shared-resource fault test.
EPS-CCF-002	Latent faults in redundant sensing, control or protection mechanisms shall be detected within the approved diagnostic interval.	Latent-fault injection and diagnostic-interval verification.
EPS-CCF-003	No single electrical/electronic fault within the allocated fail-operational scope shall propagate across independence barriers and defeat both channels.	Single-point fault matrix and containment test.

Verification, margin and future-change rule

Adding shared resources, combining connectors, changing diagnostic intervals or consolidating software/hardware functions reopens the common-cause and latent-fault analysis.

DEGRADED SINGLE-CHANNEL OPERATION & CONTROLLED RE-ENABLE

After isolation of one channel, the surviving path must use a defined degraded calibration, torque envelope and thermal strategy. Continuing normal dual-channel commands without reallocation can overheat the remaining winding or violate control assumptions. Degraded mode is therefore an engineered operating state, not only a diagnostic flag.

Re-enable is similarly controlled. A transient disappearance of the original fault is not sufficient reason to reconnect a channel that may still contain a latent inverter, sensor or processor defect. The released recovery policy must distinguish ignition-cycle recovery, service action and automatic requalification.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-DEG-001	Single-channel operation shall use a released torque, rate and thermal envelope compatible with the surviving hardware path.	Single-channel dyno envelope and thermal-duration test.
EPS-DEG-002	The EPS shall communicate degraded availability and any associated steering limitations to the vehicle.	Network-status and HMI handoff integration test.
EPS-DEG-003	A faulted channel shall not be automatically returned to full authority until the released diagnostic and recovery criteria are satisfied.	Fault-clear/re-enable negative testing.

Verification, margin and future-change rule

Changes to degraded torque limit, thermal duration, reconnection criteria or fault-latch policy require single-channel dyno and vehicle regression.

ADAS TRAJECTORY COMMAND INTERFACE & LOCAL STEERING AUTHORITY

The ADAS central compute generates trajectory demand, while the EPS retains final local authority over executable steering. This separation prevents a software or network fault upstream from directly commanding current or gate behavior. The command interface must therefore carry intent at a controlled abstraction level together with validity and timing information.

The EPS also reports actual steering state, degraded availability and local limit activation so the ADAS controller can decide whether the ODD remains valid or an MRM is necessary. Coordinate systems, sign conventions and unit scaling are safety-relevant and must be version-controlled in the ICD.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-ADIF-001	ADAS steering commands shall be accepted only when the interface validity, freshness and local-authority checks are satisfied.	Command-validity and authority negative tests.
EPS-ADIF-002	The EPS shall expose actual steering state and degraded/local-limit status needed by the vehicle automation supervisor.	ADAS/EPS closed-loop integration test.
EPS-ADIF-003	Interface version mismatch shall prevent uncontrolled command execution and shall produce diagnosable status.	Protocol-version mismatch testing.

Verification, margin and future-change rule

A change to command type, coordinate convention, status semantics, ADAS platform or MRM interface requires joint EPS/ADAS ICD and HIL regression.

CAN FD / ETHERNET, SecOC / MACsec & COMMUNICATION ROBUSTNESS

The source baseline uses secured CAN FD and Ethernet with SecOC and/or MACsec. The released vehicle topology determines which protection mechanism applies to each interface. Security services must authenticate command data without starving the deterministic steering-control and safety-supervision workloads.

Network faults include delay, duplication, reorder, burst loss, bus-off, link retraining and overload. The steering state machine must respond to the validity of the data rather than simply the physical link state. Security failure must reject untrusted commands while keeping local protection active.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-NET-001	Steering command messages shall be protected by the released freshness, integrity and authenticity mechanism before execution.	SecOC/MACsec negative authorization and replay test.
EPS-NET-002	Network overload or cryptographic processing shall not cause loss of deterministic local motor-control or safety-supervision deadlines.	Concurrent network/cyber load stress test.
EPS-NET-003	Bus-off, link loss or repeated invalid frames shall drive the allocated communication-degraded state with explicit diagnostics.	CAN FD/Ethernet fault injection.

Verification, margin and future-change rule

Network stack, MAC/PHY, gateway topology, SecOC profile, MACsec configuration or cryptographic accelerator changes require performance, fault and cyber regression.

TIMEBASE, FRESHNESS, TIMEOUT & END-TO-END DATA INTEGRITY

Trajectory execution, sensor plausibility and handover logic depend on knowing whether data is current. Each interface therefore needs a controlled freshness model based on counters, timestamps, alive supervision or equivalent mechanisms. A physically received frame is not automatically valid if it is late or replayed.

Timing budgets span sender generation, network transit, EPS task release and actuator response. The source does not release numerical network/freshness limits, so those values remain TBC until the vehicle communication and safety timing budgets are approved.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-TIM-001	Stale or replayed steering commands shall be rejected before they can refresh actuator authority.	Timestamp/alive-counter replay testing.
EPS-TIM-002	Timeout detection shall use a program-approved budget that includes worst-case sender, network and receiver timing.	End-to-end timing analysis and HIL delay sweep.
EPS-TIM-003	Loss or corruption of timebase/freshness information shall not be interpreted as valid current command data.	Timebase fault and malformed-message injection.

Verification, margin and future-change rule

Changes to scheduler, bus load, gateway path, timestamp source, time synchronization or message period require freshness and timeout regression.

THERMAL MANAGEMENT, HOT-SOAK & SAFE DERATING

The EPS dissipates losses in motor windings, inverter switches, gate drivers, power supplies and MCUs. The source baseline uses passive heat transfer to the steering gear and requires safe derating before semiconductor junction limits are violated during high-load conditions such as sustained parking or curb hold.

Thermal derating changes available steering assist and therefore interfaces with vehicle automation. The system must avoid an abrupt torque drop, report reduced capability and distinguish sensor failure from a genuine overtemperature condition. Single-channel fail-operational operation receives separate thermal treatment because load sharing changes.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-THM-001	The thermal model shall bound released motor-winding, inverter and controller hotspots across approved ambient, duty-cycle and degraded-channel conditions.	Thermal mapping and model correlation.
EPS-THM-002	Derating shall reduce steering capability progressively within the approved safety envelope and shall report the resulting availability state.	Hot-soak derating dyno/HIL test.
EPS-THM-003	Invalid thermal-sensor data shall invoke the allocated conservative strategy rather than unrestricted full assist.	Thermal-sensor fault injection.

Verification, margin and future-change rule

Motor, FET, heatsink, steering-gear interface, thermal sensor or derating-law changes require model recorrelation and hot-soak regression.

MECHANICAL RACK, PINION, HOUSING & ACTUATOR INTEGRATION

The electrical steering controller is coupled to the rack, pinion, reduction gear, bearings, seals and mounting interfaces. Mechanical tolerances affect friction, backlash, torque transfer, sensor alignment and thermal contact, so electrical safety cannot be validated independently of the released steering-gear configuration.

Mechanical binding, backlash growth, fastener looseness or gear damage can present as a control problem. Diagnostic and wear-monitoring logic must therefore avoid compensating indefinitely for a physical defect. Rack travel and hard-stop protection also require controlled interaction with motor torque limits.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-MEC-001	The released mechanical stack-up shall preserve sensor alignment, torque transfer and thermal interfaces over the approved tolerance envelope.	Tolerance analysis and build-correlation measurement.
EPS-MEC-002	Mechanical binding or abnormal friction shall be detected without allowing adaptive control to mask the condition beyond approved limits.	Binding/friction fault injection.
EPS-MEC-003	Rack travel and hard-stop conditions shall not permit sustained motor torque that exceeds the released mechanical protection strategy.	Rack-end/hard-stop dyno test.

Verification, margin and future-change rule

Rack ratio, bearing, gear, housing, mount, lubricant or mechanical supplier changes require tolerance, friction, vibration and steering-control recorrelation.

IP6K9K, VIBRATION, SALT & PRESSURE-WASH QUALIFICATION

The source baseline requires IP6K9K sealing and cites ISO 16750-3 vibration. Environmental qualification must also reflect the released installation exposure to temperature cycling, moisture, salt and pressure washing without assuming a generic vehicle location. Sealing performance must survive aging and harness/connector motion.

Sequential stress matters because vibration or thermal cycling can create leak paths that are not visible in a fresh-unit ingress test. Post-stress electrical, sensing and communication performance therefore forms part of the qualification closure rather than treating ingress as a stand-alone mechanical result.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-ENV-001	The released enclosure and connector system shall meet the approved IP6K9K sealing requirement after the applicable environmental preconditioning.	Ingress test after sequential environmental stress.
EPS-ENV-002	Post-vibration steering sensing, inverter, power and network behavior shall remain within released limits without intermittent faults.	ISO 16750-3 vibration plus functional retest.
EPS-ENV-003	Incomplete environmental qualification for a hardware or installation variant shall block release of that variant.	Environmental qualification closure review.

Verification, margin and future-change rule

Housing, gasket, vent, connector, mounting, coating or installation changes require an updated environmental matrix and post-stress electrical retest.

EMC, BCI, ESD & CONDUCTED EMISSIONS

High-current PWM switching can inject conducted and radiated noise into the vehicle network and into the EPS sensor paths. The EPS must also remain immune to external RF/BCI and electrostatic discharge without generating uncommanded steering or losing both redundant channels.

The source refers to CISPR limits but does not freeze a specific edition or vehicle-class limit. The program EMC plan therefore owns final emission/immunity levels. Control-loop operating points, PWM frequency, motor current and degraded single-channel operation must be represented in the test matrix.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-EMC-001	EPS conducted/radiated emissions shall meet the program-approved CISPR/OEM limits across representative steering load and PWM conditions.	EMC chamber emission testing.
EPS-EMC-002	BCI/radiated immunity and ESD exposure shall not produce uncommanded steering beyond the approved safety envelope.	Immunity/ESD test with torque and state monitoring.
EPS-EMC-003	EMC-induced sensor or network faults shall be detected and handled by the same validity rules used for corresponding physical faults.	EMC fault-correlation and diagnostics review.

Verification, margin and future-change rule

Harness, connector, PWM strategy, switching device, filter, grounding or PCB layout changes require targeted EMC regression based on coupling impact.

DIAGNOSTICS, DTC, FIRST-FAULT CAPTURE & SERVICE AUTHORITY

Diagnostics must support fault localization without creating a service path that can bypass steering safety. First-fault information is important in a redundant system because later secondary effects may otherwise obscure which channel or resource failed first. DTCs need stable meaning across software and hardware variants.

Service routines that exercise motor phases, sensors or calibration are permitted only in controlled states and must never obtain more actuation authority than the released service concept allows. Diagnostic communication load must also remain subordinate to motor-control timing.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-DIA-001	The EPS shall capture first-fault identity and sufficient synchronized context to distinguish primary channel failure from consequential faults.	Fault-injection log correlation.
EPS-DIA-002	Service routines capable of influencing actuation shall be state-gated and bounded by local safety limits.	Service-access negative test.
EPS-DIA-003	Diagnostic traffic shall not cause missed motor-control or safety-supervision deadlines at maximum supported service load.	Concurrent diagnostic load WCET test.

Verification, margin and future-change rule

DTC taxonomy, service tool, diagnostic stack, event snapshot or service-actuation routine changes require compatibility and safety regression.

NVM, EVENT HISTORY, THERMAL CYCLES & WEAR COUNTERS

The 15-year lifecycle concept includes thermal-cycle logging, friction-wear estimation and fault/event history. NVM must retain safety-relevant calibration and service evidence across brownouts and repeated writes without allowing corrupted history to alter steering authority.

Counters are useful only when their units, saturation, rollover, reset policy and replacement behavior are controlled. Vehicle replacement or ECU exchange also requires a defined relationship between steering-gear mechanical history and electronics history.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-NVM-001	Safety-relevant calibration and lifecycle records shall remain internally consistent across interrupted writes and brownout.	Power-loss NVM robustness test.
EPS-NVM-002	Thermal-cycle, overcurrent and friction/wear counters shall have defined units, saturation/rollover and service-reset behavior.	Counter lifecycle and boundary testing.
EPS-NVM-003	Loss of nonessential history data shall not expand steering authority or suppress a required active fault response.	Corrupted-log negative testing.

Verification, margin and future-change rule

NVM technology, wear-leveling scheme, log schema, event trigger or replacement strategy changes require endurance and power-loss consistency regression.

CYBERSECURITY, SECURE BOOT, KEY LIFECYCLE & DEBUG CONTROL

Secured vehicle communication is effective only if the EPS also protects its boot chain, software identity, credentials and maintenance interfaces. The source baseline requires SecOC/MACsec and secure OTA capability; root-of-trust implementation and key lifecycle belong to the released vehicle cybersecurity concept.

Cybersecurity mechanisms must fail in a way that preserves local steering safety. Loss of trust may cause rejection of external commands, but it must not disable local watchdogs, electrical protections or the manual/degraded steering path allocated by the safety concept.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-CYB-001	Only authenticated and integrity-verified software shall obtain released steering-control execution authority after boot.	Secure-boot tamper and rollback test.
EPS-CYB-002	Production debug/service access shall be disabled or authenticated according to the approved vehicle cybersecurity policy.	Manufacturing/debug authorization audit.
EPS-CYB-003	Cybersecurity processing or failure shall not block local safety monitoring and independent torque-protection mechanisms.	Concurrent cyber-stress and safety-fault test.

Verification, margin and future-change rule

HSM/secure element, bootloader, key hierarchy, debug policy, service authentication or crypto library changes require threat-model and concurrent safety/cyber regression.

OTA UPDATE, ROLLBACK & HARDWARE / CALIBRATION COMPATIBILITY

The source lifecycle includes secure OTA. Steering software update must be atomic with respect to safety calibration, network configuration and hardware compatibility so that a partially updated ECU cannot boot with mismatched current limits, sensor scaling or fail-operational logic.

Recovery behavior depends on the vehicle power architecture and bootloader design. The Master PRD requires at least one validated recovery path and explicit compatibility checks before new software is given actuation authority. Update failure is a release/service condition; it must not silently enable an unqualified degraded configuration.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-OTA-001	Interrupted OTA shall preserve or recover to an authenticated software/calibration combination approved for the installed EPS hardware.	Power-fail update and recovery test.
EPS-OTA-002	Software shall reject incompatible hardware, calibration or network configuration before granting steering actuation authority.	Variant-mismatch negative testing.
EPS-OTA-003	Rollback policy shall prevent return to a software version that is revoked by the approved cybersecurity/safety baseline.	Anti-rollback and recovery authorization test.

Verification, margin and future-change rule

Bootloader, partitioning, signing policy, calibration format, ECU hardware variant or update transport changes require power-fail and compatibility regression.

MANUFACTURING EOL, PROGRAMMING & PRODUCT GENEALOGY

Production must prove that the assembled EPS corresponds to the released hardware, software, calibration and security identity. EOL programming alone is not sufficient; manufacturing records must link motor, inverter, sensing, controller and steering-gear variants to the acceptance evidence used for that unit.

The source requires EOL sensor offsets, friction maps and cryptographic keys. Fixture uncertainty and process capability must be included so an apparent pass is not created by a poorly controlled measurement system. Runtime software trim cannot be used to hide an incapable EOL process.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-MFG-001	Each production EPS shall be traceable to released hardware, software, calibration, security credentials and critical component genealogy.	Manufacturing genealogy audit.
EPS-MFG-002	EOL acceptance measurements shall use traceable fixtures with demonstrated measurement-system capability.	Gauge R&R and fixture correlation.
EPS-MFG-003	A unit with incomplete EOL evidence or unresolved genealogy shall not be released as production-equivalent.	EOL release-gate audit.

Verification, margin and future-change rule

Assembly process, fixture, programming station, key-provisioning flow, supplier or acceptance-limit changes require manufacturing validation and genealogy review.

SENSOR CALIBRATION, RACK CENTER & FRICTION-MAP ALIGNMENT

EOL calibration aligns torque/angle sensing, motor position, rack center and friction compensation to the assembled steering gear. These values are interdependent: a rack-center error can appear as trajectory tracking error, while sensor offset can bias both manual assist and autonomous torque.

Calibration limits and fixture methods remain program-controlled. The important Master PRD rule is that calibration cannot compensate an out-of-spec mechanical assembly or sensor. Residuals beyond acceptance must reject or rework the unit rather than expanding software trim range.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-CAL-001	Torque/angle and motor-position calibration shall produce residual error within the released acceptance limits using traceable references.	EOL calibration correlation.
EPS-CAL-002	Rack-center calibration shall be consistent with the vehicle steering reference and shall not be used to mask an assembly misalignment beyond approved tolerance.	Rack-center fixture and vehicle alignment correlation.
EPS-CAL-003	Friction-map generation shall reject abnormal binding or assembly defects rather than learning them as normal compensation.	Faulted-mechanics calibration negative test.

Verification, margin and future-change rule

Sensor supplier, target geometry, rack assembly, fixture, calibration algorithm or friction-map method changes require calibration-correlation and vehicle-center verification.

HARDWARE DVP - POWER, INVERTER, MOTOR & SENSOR CHAINS

Hardware DVP demonstrates the physical steering channels across supply, temperature, load, fault and component tolerance. Tests must cover normal dual-channel operation and the surviving single-channel state because the latter changes thermal loading, torque capability and fault exposure.

Power-stage short circuit, open phase, current-sensor faults, motor-position faults, TAS faults, brownout and source isolation are minimum categories derived from the source architecture. Exact sample count and acceptance values remain program-controlled and must be frozen in the DVP matrix.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-HDVP-001	Hardware DVP shall cover the released normal and degraded steering envelopes with objective pass/fail criteria.	Configuration-controlled hardware DVP matrix.
EPS-HDVP-002	Fault injection shall include inverter, power, motor-position, TAS and current-sensing failures with synchronized torque/state measurement.	Instrumented bench/dyno fault campaign.
EPS-HDVP-003	Unclosed high-severity hardware DVP failures shall block equivalence of the affected hardware variant.	DVP closure review.

Verification, margin and future-change rule

Hardware architecture, critical component, supplier or acceptance-limit changes require targeted regression proportional to the affected failure mechanisms.

CONTROL / DYNO DVP - MANUAL, L3 & DEGRADED STEERING

Control DVP verifies torque accuracy, angle tracking, stability, steering feel, handover and degraded single-channel performance on a representative rack/motor dynamometer. The objective is to expose interactions that cannot be seen in software-only simulation, including friction, saturation, backlash and thermal derating.

The source requires no torque discontinuity across mode transitions and retention of fail-operational capability. Tests therefore span manual assist, L3 command tracking, driver override, limit entry/recovery, channel isolation and recovery behavior across the released load and speed envelope.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-CDVP-001	Dyno DVP shall verify manual-assist and autonomous-command behavior across the released speed/load and temperature envelope.	Instrumented rack/motor dyno matrix.
EPS-CDVP-002	Mode transition, driver override and degraded-channel entry shall be assessed for torque discontinuity and controllability.	Transition torque-ripple and response testing.
EPS-CDVP-003	Closed-loop stability and saturation recovery shall be demonstrated with representative mechanical tolerance and friction conditions.	Plant-variation robustness dyno test.

Verification, margin and future-change rule

Control gains, rack mechanics, motor variant, assist maps, blending logic or degraded calibration changes require dyno regression before vehicle release.

HIL / VEHICLE VALIDATION - LANE CHANGE, EVASIVE & MRM SCENARIOS

The source explicitly calls for vehicle-level simulation and fault injection during L3 maneuvers such as lane changes and evasive steering. HIL provides repeatable network, power and controller faults, while vehicle testing confirms the mechanical/driver interaction and actual vehicle-dynamics consequences.

Scenario validation must include command loss, stale data, sensor faults, channel isolation, brownout, thermal derating and MRM coordination. The EPS pass/fail criteria are its allocated steering behavior and timing; broader ODD success remains the parent ADAS/vehicle program responsibility.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-HIL-001	HIL shall inject representative EPS, network and power faults during L3 steering scenarios with objective timing and torque acceptance limits.	Scenario-based HIL fault matrix.
EPS-HIL-002	Vehicle validation shall confirm driver controllability and steering response for manual handover and degraded/MRM-support states.	Instrumented proving-ground / vehicle test.
EPS-HIL-003	Recorded scenario evidence shall identify software, calibration, hardware and vehicle configuration used for each result.	Scenario genealogy and evidence audit.

Verification, margin and future-change rule

Changes to ADAS interface, vehicle dynamics, rack ratio, handover concept or MRM strategy require joint scenario regression.

SAFETY FAULT INJECTION, FTTI & INDEPENDENCE VALIDATION

Safety validation closes the end-to-end claim that a credible fault is detected, contained and moved to the allocated steering state within the approved timing budget. It also demonstrates that independent protection remains effective when the primary controller or communication path is intentionally compromised.

Tests are selected from the safety analysis rather than from a generic fault list. Common-cause, latent, delayed-detection and plausible-wrong faults require attention because they can evade nominal diagnostics. Missing evidence is a release blocker, not a runtime fallback concept.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-SVAL-001	Safety fault injection shall trace each tested fault to the corresponding safety mechanism and expected steering state.	Safety verification matrix and traceability review.
EPS-SVAL-002	Measured end-to-end response shall demonstrate closure of the approved FTTI allocation under worst-case relevant conditions.	Synchronized fault-to-torque timing capture.
EPS-SVAL-003	Independence claims shall be tested by disabling or corrupting the primary path while exercising the allocated backup/protection path.	Processor/network/protection independence tests.

Verification, margin and future-change rule

Safety mechanism, FTTI allocation, diagnostic interval, external mitigation or independence argument changes require targeted safety regression.

15-YEAR RELIABILITY, WEAR & PREDICTIVE MAINTENANCE

The source defines a 15-year vehicle lifecycle with thermal-cycle logging, friction-wear estimation and secure OTA capability. Reliability engineering must connect that mission to component stress, motor/inverter aging, sensor drift, connector degradation, NVM endurance and steering-gear wear without assuming all vehicles see the same duty cycle.

Predictive indicators are advisory until correlated with physical failure mechanisms. A friction trend or thermal-cycle count may trigger service or inspection, but it cannot automatically expand operating limits or suppress an active safety fault. Replacement of the ECU versus complete steering gear also requires explicit history transfer rules.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-LIFE-001	Lifecycle analysis shall relate the approved vehicle mission profile to motor, inverter, sensor, connector and NVM aging mechanisms.	Mission-profile stress and reliability review.
EPS-LIFE-002	Wear/thermal indicators shall be correlated to physical degradation before they are used for service thresholds.	Accelerated aging and field-correlation evidence.
EPS-LIFE-003	A lifecycle threshold exceedance shall not silently alter safety limits; the resulting service/degraded action shall be explicitly defined.	Threshold and service-action testing.

Verification, margin and future-change rule

Mission profile, component life model, wear threshold, service strategy or lifetime extension changes require renewed reliability analysis and accelerated correlation.

SUPPLIER CHANGE, SECOND SOURCE & TECHNOLOGY REFRESH

MCUs, power semiconductors, sensors, connectors and passive components may become obsolete during the 15-year support period. A form-fit-function statement alone is insufficient for steering because subtle changes in timing, thermal impedance, sensor transfer function or EMC behavior can invalidate prior evidence.

Second-source approval therefore uses similarity analysis to identify which WCCA, safety analysis, software calibration and DVP evidence can be reused and which must be repeated. Production equivalence is an engineering release decision, not a procurement-only decision.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-SUP-001	A second-source component shall not be approved until documented similarity analysis identifies affected safety, electrical, thermal, EMC and calibration evidence.	Supplier change-impact review.
EPS-SUP-002	Required targeted regression shall be closed before the changed component is granted production equivalence.	Regression DVP closure audit.
EPS-SUP-003	Supplier changes affecting hidden firmware or calibration shall be configuration controlled with the same rigor as hardware changes.	Supplier firmware/configuration audit.

Verification, margin and future-change rule

Any supplier, die revision, package, manufacturing site, material or critical firmware change enters formal change-impact review before equivalence is claimed.

STANDARDS APPLICABILITY, TBC REGISTER & EVIDENCE OWNERSHIP

The source explicitly references AUTOSAR CP, SecOC/MACsec, ASIL-D intent, IP6K9K, ISO 16750-3 and CISPR limits. Exact standard editions, OEM specifications and regulatory applicability must be approved by the program. This page therefore separates source-defined technology from candidate frameworks that require formal applicability decisions.

Candidate program frameworks may include ISO 26262 for functional safety, ISO/SAE 21434 for cybersecurity, CISPR 25 / OEM EMC methods, ISO 11452 immunity, ISO 7637 transients and applicable steering regulations. These are not compliance claims in this review baseline. The standards matrix must identify owner, edition, tailoring and evidence location.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-STD-001	The program shall maintain an approved standards/applicability matrix identifying edition, scope, owner and required evidence for the EPS release.	Standards applicability review.
EPS-STD-002	All open numerical parameters shall be recorded in a controlled TBC register with owner, rationale, due gate and affected verification.	TBC register audit.
EPS-STD-003	No candidate standard, example limit or planning value shall be represented as a released compliance claim without approved evidence.	Document claim and evidence audit.

Verification, margin and future-change rule

A change in market, vehicle platform, installation, standard edition or OEM requirement requires applicability review and can reopen affected qualification evidence.

FINAL RELEASE CLOSURE & CONFIGURATION FREEZE

Production release closes the Master PRD only when the approved product baseline, L2 subsystem PRDs, software/calibration identities, safety evidence, cybersecurity evidence, environmental qualification, manufacturing capability and DVP results form one traceable configuration. A narrative design review cannot substitute for missing objective evidence.

The release package must also identify residual TBCs, approved deviations and field/service constraints. A configuration can be released only for the variants actually covered by evidence; similarity to another steering gear, vehicle power architecture or ADAS interface is not assumed without documented impact analysis.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
EPS-REL-001	Every atomic Master PRD requirement shall trace to an approved L2 allocation and objective verification or analysis evidence.	Final requirements/evidence traceability audit.
EPS-REL-002	The released hardware, software, calibration, manufacturing and supplier configuration shall be uniquely identifiable and reproducible.	Configuration freeze and genealogy review.
EPS-REL-003	Open safety-critical TBCs, unresolved high-severity DVP failures or incomplete qualification evidence shall prevent final production release.	Final release-board closure audit.

Verification, margin and future-change rule

After release, changes follow formal configuration control with explicit regression scope. The final release baseline date remains Program Date TBC until approved by the responsible program authority.